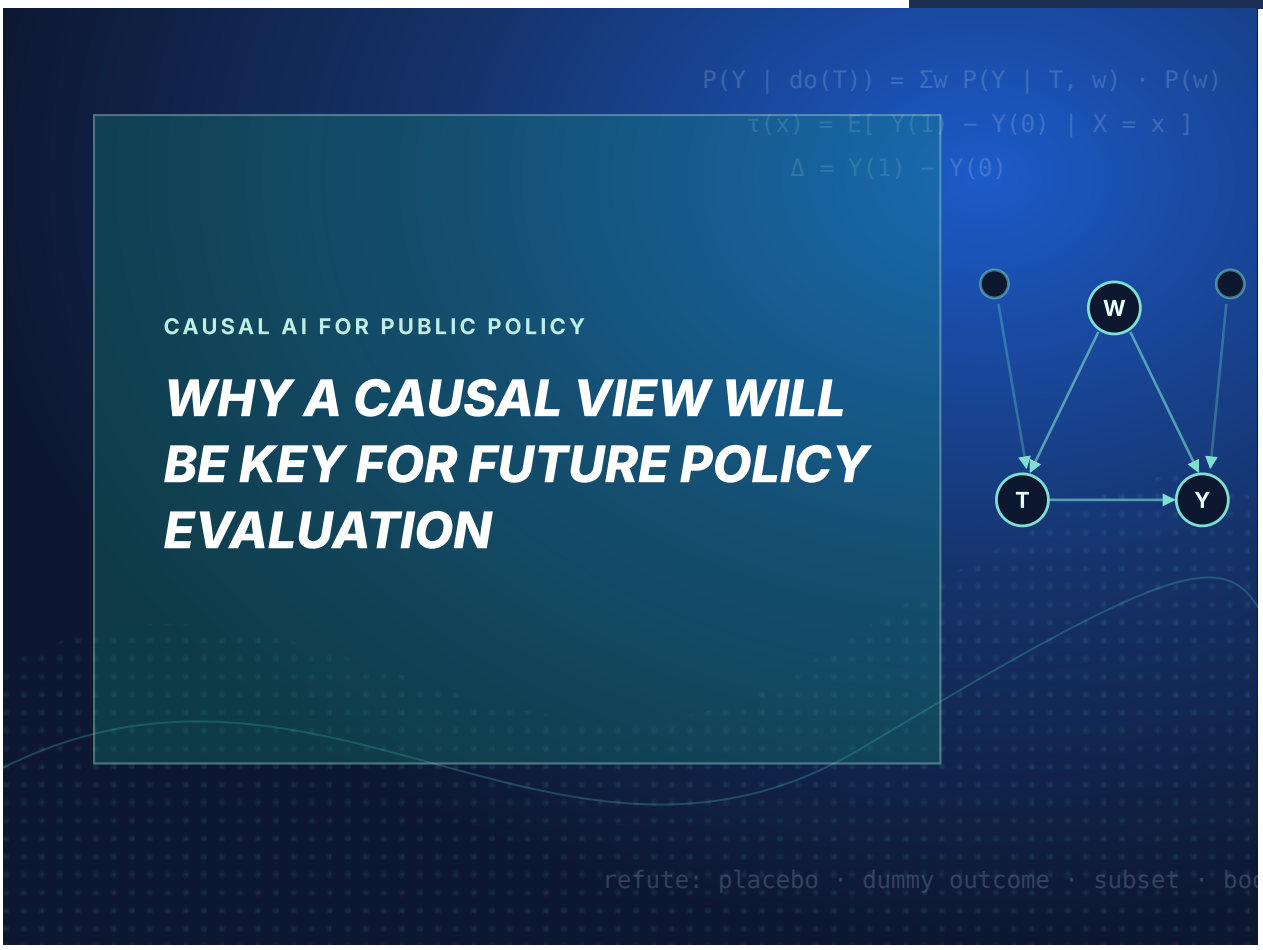
Decision intelligence,
powered by causality.

EVA

WHITE PAPER



CAUSAL AI FOR PUBLIC POLICY

WHY A CAUSAL VIEW WILL BE KEY FOR FUTURE POLICY EVALUATION

FROM CORRELATION STATISTICS TO "CAUSAL POLICY ANALYTICS"

Evidence base

Proof of concept built for the Korea Small Business Institute (KOSI), AI+ OpenData Challenge, Task 4 "Policy Effect Prediction Analysis", 2026

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SUMMARY

Public support programs are among the largest investment decisions a society makes. Yet in most agencies, the effect of a program is still judged by **beneficiary statistics**: how did the sales or headcount of supported firms develop? Those statistics are perfect for what they were built for – accounting for where the money went. They were not built to answer the question decision-makers actually ask: *did the program cause the change?*

The difference matters because firms do not receive support at random. They apply, they are screened and they are selected. The firms that end up in a program are, by construction, different from those that do not – and that difference, not the program, may drive the outcome.

A real example from this paper: in Korea's SME support data (7.92 million rows of firm records, 4.42 million support records, 2,035,733 firms), supported firms grew *more slowly* than comparable unsupported firms. A correlation dashboard would conclude that support "does not work". A causal analysis asks *why* – and finds that roughly 89% of participation is concentrated in finance-type support (guarantees and loans) that is, by design, allocated to firms already in difficulty.

That is what **Causal Policy Analytics** facilitates. Instead of asking "how did participants perform?", it asks "how would the same firms have performed without the program?" – `P(Y | do(T))` rather than `P(Y | T)`. It makes causal assumptions explicit in a directed acyclic graph (DAG), reduces selection bias on observed characteristics by matching, estimates effects firm by firm (conditional average treatment effect, CATE), tries to break its own result with refutation tests, and only then forecasts a program that does not exist yet.

This White Paper explains the paradigm and the five-level pipeline that EVA built and validated as a proof of concept (PoC) for the Korea Small Business Institute (KOSI) in the AI+ OpenData Challenge: from parsing policy documents to a Korean-language decision report. Four refutation tests passed in both cohorts, firm-level CATE error stayed between 2.6% and 11.9% against a 15% target, and every number in the report is computed by the causal engine – never generated by the language model.

You will learn why understanding cause and effect, not only correlation, will be the key element of future policy AI – and why such a system must also be honest about what observational data *cannot* identify.

2,035,733
firms in the service layer, 210 programs, 833 notices – none dropped

4 / 4
DoWhy refutation tests passed in the 2022 and 2023 cohorts (target 3 / 4)

2.6–11.9%
firm-level CATE error, 8 of 8 outcome–cohort combinations within the 15% target

−3.56%p
sales-growth estimate (2022 cohort) – and the reason it must not be read as "harm"

CAUSAL POLICY ANALYTICS VS. CORRELATION STATISTICS

WHAT BENEFICIARY STATISTICS ARE MEANT FOR – AND WHAT NOT

The first step of a classic program evaluation is to collect the outcomes of the firms that were supported: sales, employment, survival, patents. This is fast and transparent, and it is very generic – no assumptions about why a firm was supported need to be made.

But in almost every policy question, a specific **comparison** is the focus of attention: the supported firm versus the *same* firm without support. That second firm is never observed. Whatever we put in its place – all unsupported firms, last year's average, the industry trend – carries its own differences into the result.

Whenever selection into a program is not random, a raw comparison mixes two things that look identical in a table: the effect of the program and the effect of who was chosen.

HOW CAUSAL POLICY ANALYTICS DIFFERS

The business plan behind our PoC traced the problems of conventional policy analysis to its **method design**. Six structural gaps recur (Figure 1). Apart from document handling, which L1 automates, Causal Policy Analytics closes them with five testable steps:

1. state the causal assumptions in a DAG and identify the effect with the backdoor criterion;
2. build a quasi-experiment by matching (PSM[1], CEM[2]);
3. estimate heterogeneous, firm-level effects (CATE);
4. attack the estimate with refutation tests;
5. derive counterfactuals for programs not yet run.

Analytical capability	Conventional statistics	ML prediction	Causal Policy Analytics
Detect correlation	Yes	Yes	Yes
Identify causation	No	No	**Yes – DAG + backdoor criterion**
Remove selection bias	Partial	No	**On observed confounders – PSM / CEM matching**
Firm-level heterogeneous effect	No	No	**Yes – CATE (Causal Forest)**
Counterfactual ex-ante prediction	No	No	**Yes – counterfactual Y1 = Y0 + CATE**
Automatic validation of the estimate	Manual	No	**Automated – DoWhy refutation tests**
Explainability	High	Low	**High – explicit causal paths**

Figure 1 – Capability comparison of conventional statistics, machine-learning prediction and Causal Policy Analytics. The six gaps – manual document handling, correlation read as causation, uncontrolled selection, assumed effect homogeneity, missing counterfactuals and black-box results – were defined in EVA's KOSI business plan.

FROM P(Y | T) TO P(Y | DO(T))

Judea Pearl's do-calculus[3] gives the distinction a precise form. `P(Y | T)` describes what we *observe* among firms that happened to participate. `P(Y | do(T))` describes what would happen if we *intervened* and assigned participation. The two are guaranteed to coincide only when no confounder opens a backdoor path between participation and outcome.

In SME policy, many things do: industry, size, region, age, legal form, patents, financial health, headcount and the firm's own growth trend. These **confounders** (W) open a "backdoor" path T ← W → Y. If the DAG is right and W blocks every backdoor path (Figure 2), the interventional effect can be computed from observational data:

$$P(Y \mid do(T)) = \Sigma_w P(Y \mid T, w) \cdot P(w)$$

This is the identification step. It does not estimate anything yet; it decides *which* comparison is valid. Everything downstream – matching, CATE, counterfactuals – inherits its credibility from this step.

ONE QUESTION, ONE ESTIMAND

A second, often overlooked consequence: different policy questions require different causal quantities. Our service deliberately shows different estimands on different screens (Figure 3). An ex-post evaluation asks about the firms that were actually treated (average treatment effect on the treated, ATT). A recommendation for a firm that has never participated asks about firms *like it* (group average treatment effect, GATE, or CATE). The population-wide average is the ATE. Reporting one "average effect" for all three questions would be precise and wrong.

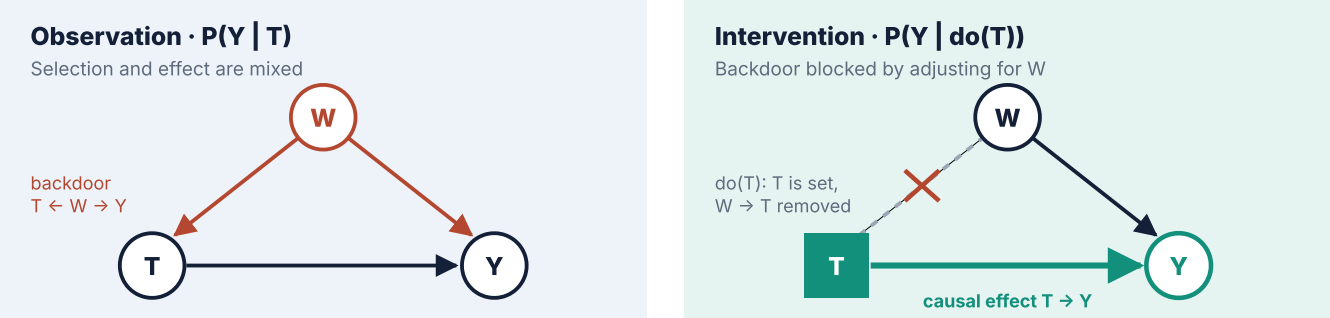


Figure 2 – Observation versus intervention. Left: confounders W influence both participation T and outcome Y, so a raw comparison mixes selection with effect. Right: adjusting for W along the backdoor criterion isolates the causal path T → Y.

Screen / situation	Estimand	Why this estimand
One firm selected (menu 1 results)	**CATE(x)**	Conditional effect predicted from the firm's covariates X – not an individual treatment effect: two firms with identical X receive the same value.
Menu 4 "effect of this program"	**ATT**	Average over the program's actual participants (T=1); ex-post evaluation naturally refers to the treated group.
Menus 1, 2, 6 "expected effect"	**GATE**	No actual participants exist yet; group average of firms matched on industry, size and other covariates (matched regardless of treatment status).
Population figure (methodology tab)	**ATE**	Model applied to the whole cohort (T=1 and T=0), e.g. −3.25%p – the only "textbook" ATE among these figures.

Figure 3 – Matching the estimand to the question (selected screens). Evaluation (what happened) calls for ATT; recommendation and design (what would happen) call for GATE or CATE.

THE FIVE-LEVEL CAUSAL PIPELINE

Causal estimates are only as good as the data that feed them, and in public policy much of that data is not in a table. Program conditions, eligibility and budgets live in PDF and HWP notices. EVA's pipeline therefore takes **two inputs** – unstructured documents and structured firm records – and joins them in an ontology knowledge graph before any effect is estimated (Figure 4).

L1 parses notices into a Document AST (JSON) and indexes them for retrieval. **L2** turns firm records into treatment (T), outcome (Y) and confounder (W) variables. **L3** fixes the DAG and builds matched treatment and control cohorts. **L4** estimates firm-level effects with EconML's CausalForestDML. **L5** derives the counterfactual and writes the report.

The knowledge graph (Neo4j) links firm, program, support history and outcome indicators. It supplies candidate confounders for the L3 DAG and multi-hop context for retrieval: 3,593 nodes and 24,293 edges in the document graph; firm nodes (2,035,733) are loaded in a Tier-2 ETL of 3.8 million edges.

The levels are not interchangeable. Without identification, matching is meaningless; without validation, no estimate may be used as policy evidence.

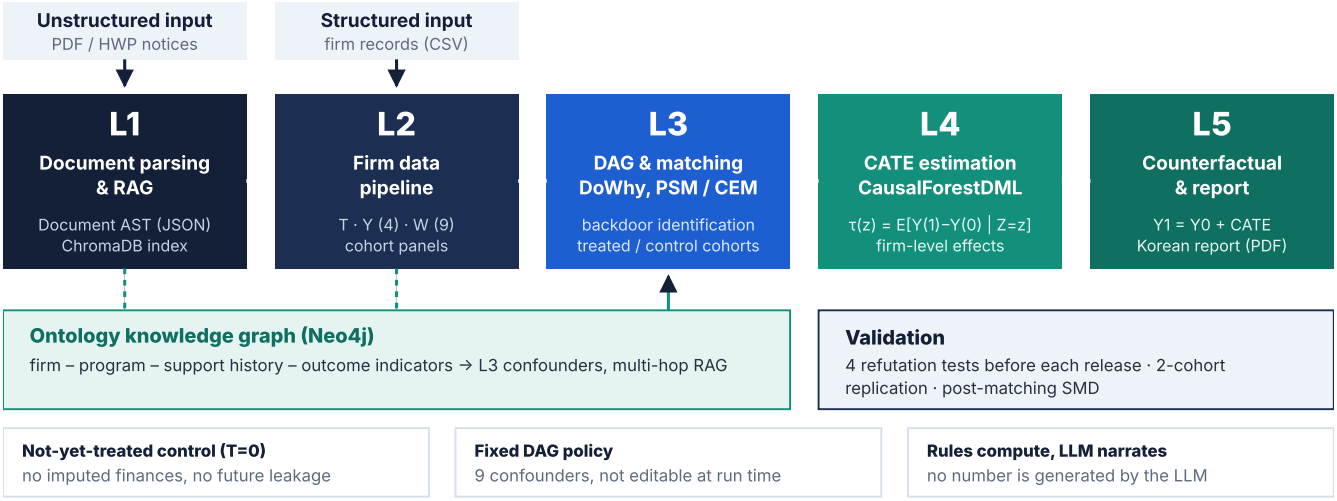


Figure 4 – End-to-end architecture of the PoC: unstructured and structured inputs are joined in an ontology knowledge graph, causally identified (L3), estimated (L4) and turned into a counterfactual report (L5). Bottom row: the three design principles.

THREE DESIGN PRINCIPLES

Cohort-based control group. A firm is treated (T=1) if its first support falls in the base year, and control (T=0) if it remains unsupported until the following year. The status is decided only by cross-checking two independent tables – financial ledger and support history – so no financial information is assumed or generated and no future information leaks into the comparison.

Fixed DAG. The nine confounders are reviewed by domain experts and written into a single module (`dag_builder.py`). They cannot be altered ad hoc at evaluation time, so the identifying assumptions are auditable.

Rules compute, the language model narrates. CATE, Y0, Y1 and the confidence grade are produced by levels L1–L4 and inserted unchanged. The LLM (self-hosted Gemma 4, 31B) only fills three narrative slots – context, selection bias and implications – which removes numerical hallucination by construction.

EXAMPLES FOR CAUSAL POLICY ANALYTICS

DEFINING THE CAUSAL MODEL: T, Y AND NINE CONFOUNDERS

Our PoC used the data KOSI provided for the challenge: firm basic and financial records (7,921,066 rows, 45 columns), support history (4.42 million records, 32 columns), 210 programs, 833 notices and two satisfaction surveys. EVA added 1,977 original notice documents (1.1 GB) so that program conditions could be read from the source text.

From these, L2 builds one variable set per cohort (Figure 5) and the matched quasi-experiment (Figure 6). **T** is program participation (0/1) under the not-yet-treated design. **Y** comprises four outcomes, each with its own model: sales growth, employment growth, closure and change in patents. **W** comprises nine confounders measured only before treatment.

Sales, exports and employment are recorded as bands ("below KRW 0.5 bn", "0.5–1 bn"). We convert each band to its arithmetic midpoint, (upper + lower) / 2; bands without a lower bound use upper / 2, bands without an upper bound use lower × 1.5. A log-midpoint alternative differs by 5.7–9.7% and is reported alongside. R&D expenditure was excluded because 95.47% of values fall in the "other" category.

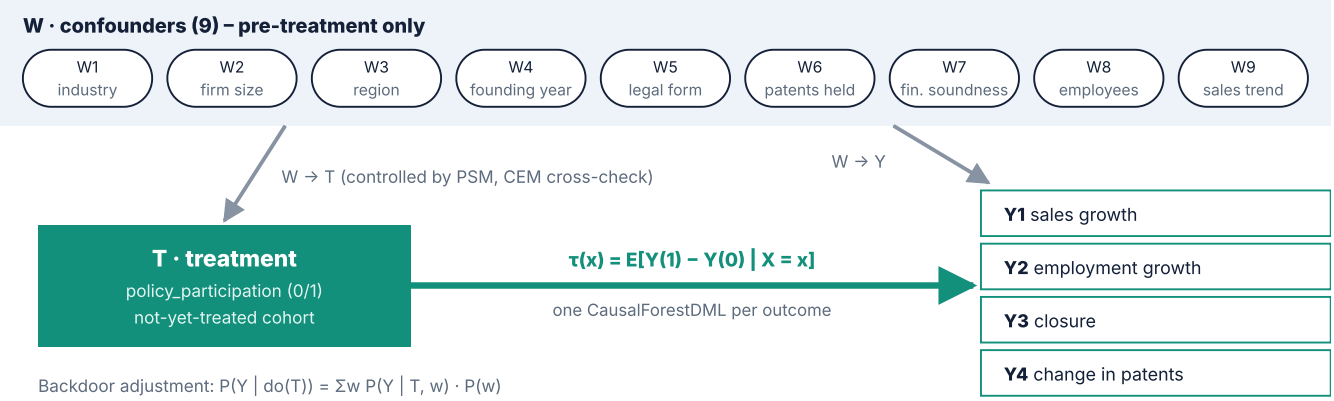
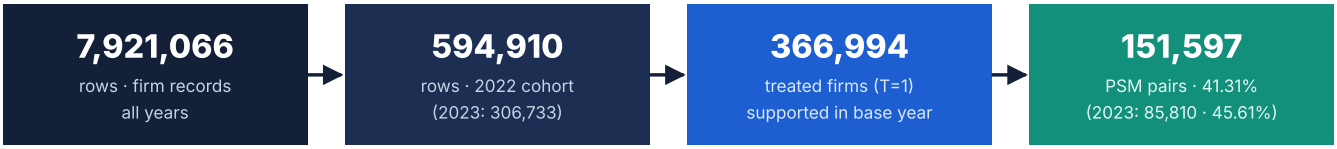


Figure 5 – The fixed DAG used in the PoC (nine confounders finalised on 8 August 2026). Confounding paths W → T, Y are controlled by propensity-score matching with a CEM cross-check; each of the four outcomes has its own CATE model.



The service layer keeps all 2,035,733 firms; the trained model is applied to every firm to produce an individual effect estimate.

Figure 6 – From raw records to a matched quasi-experiment (2022 cohort; 2023 values in brackets). Learning uses refined cohorts; prediction covers the full population.

PROGRAMMING INTERFACES, E.G. PYTHON

All five levels are built on open-source components; the causal and machine-learning core runs on Python libraries under MIT, Apache 2.0 and BSD-3 licences. The causal core uses two Microsoft Research projects: **DoWhy**[4] for modelling, identification and refutation, and **EconML**[5] for heterogeneous effect estimation.

Figure 7 shows, in simplified form, how the causal model is declared and identified. The declaration records the assumptions, and identification derives the estimand symbolically – no estimation is performed yet. The heavy lifting starts only when an estimator is called.

Figure 8 shows the estimation step. CausalForestDML combines double/debiased machine learning[6] with the causal forest of Wager & Athey[7]: nuisance models for outcome and treatment are fitted first (gradient boosting in our PoC), then a forest splits on covariates where the treatment effect differs. A minimum of 50 samples per leaf, set when the model was retrained on 22 July 2026, keeps each local estimate stable; out-of-bag estimation prevents overfitting.

The result is not a single number but a distribution of effects with confidence intervals – the basis for every recommendation, comparison and program design in the service.

```
from dowhy import CausalModel

W_VARS = ["industry", "firm_size", "region", "founding_year", "legal_form",
          "patents_held", "fin_soundness", "employees", "sales_trend"]   # fixed DAG

model = CausalModel(
    data=cohort_2022,                      # not-yet-treated cohort panel (L2)
    treatment="policy_participation",      # T
    outcome="sales_growth",                # one of four Y
    common_causes=W_VARS)                  # W

estimand = model.identify_effect()         # backdoor: P(Y|do(T)) = Σw P(Y|T,w)·P(w)
```

Figure 7 – Declaring and identifying the causal model with DoWhy (simplified illustration of the public API; variable names abbreviated, not production code). Identification is symbolic – no estimation is performed yet.

```
from econml.dml import CausalForestDML
from sklearn.ensemble import GradientBoostingRegressor, GradientBoostingClassifier

est = CausalForestDML(
    model_y=GradientBoostingRegressor(),   # nuisance model for Y
    model_t=GradientBoostingClassifier(),  # nuisance model for T (propensity)
    discrete_treatment=True,
    min_samples_leaf=50)                   # retrained 2026-07-22
est.fit(Y, T, X=X_firm, W=W_ctrl)

cate   = est.effect(X_new)                  # τ(x) = E[Y(1) − Y(0) | X = x] per firm
lb, ub = est.effect_interval(X_new, alpha=0.05)   # 95% CI → error metric
y1     = y0_matched_mean + cate             # counterfactual headline: Y1 = Y0 + CATE (rule layer)
```

Figure 8 – Firm-level effect estimation with EconML (simplified illustration). Y0 is the observed mean of matched non-participants; the rule layer – not the language model – adds the CATE to obtain the counterfactual figure shown in the report.

TRYING TO BREAK THE RESULT: VALIDATION

A causal estimate is a claim, and a claim must survive attempts to falsify it. In the business plan we fixed the acceptance criterion *before* development: at least three of four DoWhy refutation tests must pass (planned set: placebo treatment, random common cause, data subset, bootstrap). In the PoC, the executed set – placebo treatment, dummy outcome, data subset and bootstrap – passed four of four in both cohorts (Figure 9). It runs automatically before every release, together with 985 automated pytest checks.

The logic of each test is simple. Replace the real treatment with a random one and the effect must vanish (placebo). Replace the outcome with noise and no effect may appear (dummy outcome). Remove 20% of the data and the conclusion must hold (subset). Resample 100 times and the estimate must stay within a stable range (bootstrap).

Accuracy is measured separately. The CATE error rate – based on the mean width of the firm-level 95% confidence interval under 5-fold cross-validation – stayed between 2.6% and 11.9% for all eight combinations of four outcomes and two cohorts. Covariate balance after matching was checked with the standardised mean difference (SMD) on all nine confounders.

Finally, reproducibility: the 2022 cohort was the main analysis and the 2023 cohort an independent replication. The sales-growth estimate points in the same direction in both (−3.56%p and −2.82%p), and three different estimators agree on the sign (Figure 10).

Area	Check	Method	Criterion	Result
Refutation	Placebo treatment	Treatment replaced by a random variable, re-estimated	Effect → 0	**PASS**
	Dummy outcome	Outcome replaced by a random variable	No effect detected	**PASS**
	Data subset	20% of data removed, re-estimated	Same conclusion	**PASS**
	Bootstrap	100 resamples, variance of estimate	Consistent result	**PASS**
Accuracy	CATE error rate	Mean width of firm-level 95% CI, 5-fold CV	≤ 15%	**2.6–11.9% (8/8)**
Balance	Covariate balance	Post-PSM SMD on nine confounders	SMD threshold	**Met**
Replication	Cohort cross-check	2022 main vs. 2023 independent re-analysis	Same direction	**−3.56 / −2.82%p**

Figure 9 – Validation results of the PoC. Refutation tests run before every release; the business plan required 3 of 4 (its planned set used a random-common-cause test instead of the dummy outcome).

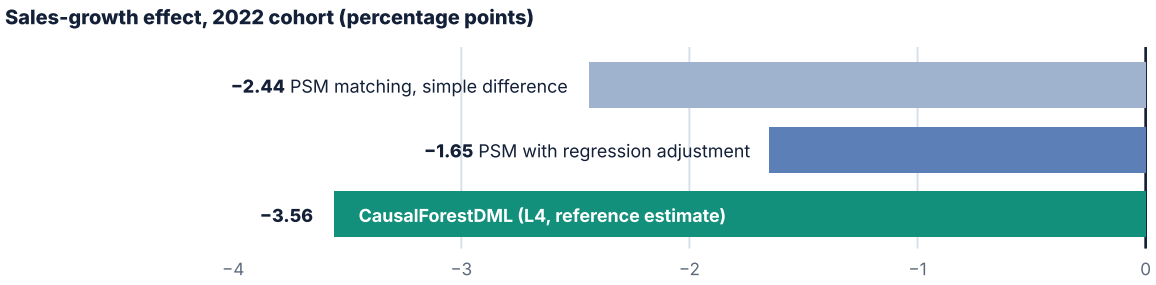


Figure 10 – Three estimators, one sign. All three point in the negative direction, and the result is reproduced in the 2023 cohort. The next pages explain why this negative number must not be read as “support harms firms”.

CASE STUDY: WHEN "NO EFFECT" REALLY MEANS "SELECTION"

Consider a simple observation: firms that take emergency loans grow more slowly than firms that do not. Do emergency loans slow firms down? Or do slowing firms take emergency loans?

The correlation between "receiving support" and "lower growth" is real. Its interpretation is not obvious. A beneficiary report would show a negative figure and invite the conclusion that the policy failed. This is exactly the point where, in the PoC data, a correlation analysis would have stopped (Figure 11).

A causal workflow does not stop at the number. It asks why, and it treats the anomaly as a finding about the data-generating process. Three pieces of evidence emerged.

1. Pre-trends are balanced. Before support (2020→2021), the raw gap in sales growth between participants and non-participants was −1.6%p; within strata it practically disappears (+0.46%p). The comparison groups looked alike before treatment.

2. The post-period gap persists. After support (2021→2023), the raw gap of −2.1%p becomes −2.9%p within strata (finance and domestic-market types) – matching on observed characteristics does not close it (Figure 12).

3. Matching success is itself a signal. For finance-type support, only 36–42% of participants found a matched twin, against 98% for domestic-market support. The finance group is so distinctive that observable characteristics cannot reproduce it in the control pool.

Outcome	2022 cohort	2023 cohort	Interpretation
Sales growth	−3.56%p	−2.82%p	Driven by finance-type support (share ~89%); same direction, replicated
Employment growth	−0.97%p	−0.46%p	Same direction as sales, smaller magnitude
Closure	+12.75%p	+8.87%p	Higher closure probability among participants – same root cause: crisis-response support
Change in patents	+0.21%p	+0.18%p	Negligible short-term effect; effect expected after 3–5 years

Figure 11 – Estimated effects by outcome and cohort after matching (CausalForestDML). Read naively, the table says "support reduces growth and increases closure".

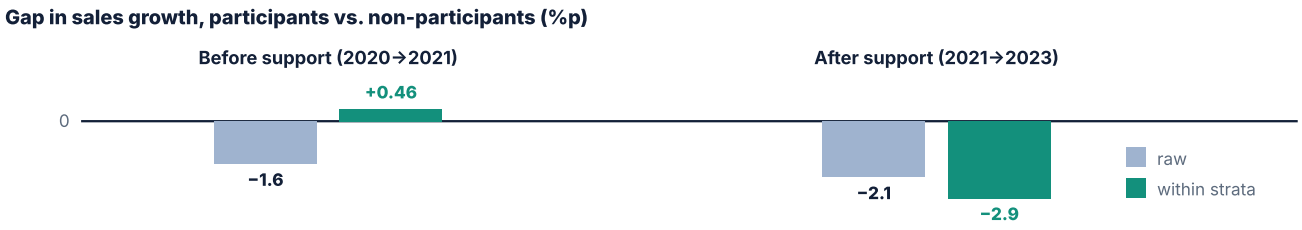


Figure 12 – Stratification removes the pre-treatment gap but not the post-treatment gap (post-period: finance and domestic-market types).

THE ROOT CAUSE: 89% CRISIS-RESPONSE SUPPORT

The composition of participation explains all three observations at once (Figure 13). In both cohorts, 88.8% and 89.7% of participation falls on finance-type support: guarantees and loans such as general and special management-stabilisation funds, support for SMEs in temporary distress or disaster, restart funds, restructuring funds and business-conversion funds.

Such programs are, by design, allocated to firms that are already in difficulty. Firms that are doing well tend not to apply. The lower average performance of this group is therefore most plausibly explained by **who was selected** rather than by the support itself. A plausible mechanism consistent with the evidence: a firm applies because of a shock that is not yet visible in its past figures – which is why balanced pre-trends do not rule out selection, and why the closure estimate turns positive.

The same analysis also shows where the reading is more robust. For smart-manufacturing and AI programs (76,206 support cases), the finance-type share is only 4.3% (versus 88.8% overall) and the domestic-market share 23.8% (versus 3.7%). The results for this program group are far less exposed to crisis-response selection.

> ***A correlation analysis would have reported "no policy effect". A causal analysis reports what the data can and cannot identify – and where to look next.***

Figure 14 contrasts the readings. We deliberately publish the negative figure rather than hiding it. Its value lies in the diagnosis: observed confounders W1–W9 cannot fully remove selection into crisis-response support. The final section describes the data needed to close that gap.

Support type (program area)	2022 cohort	2023 cohort	Matching and effect status
Finance (guarantee, loan)	**88.8%**	**89.7%**	Matching success 36–42%; drives the negative estimate
Other (management etc.)	7.5%	6.6%	Only 2022 reliable (2023 pre-trend imbalance) – interpretation withheld
Domestic market expansion	3.7%	3.7%	Matching success 98%; direction not consistent (2022 negative, 2023 not significant)

Figure 13 – Share of participation by support type. Example finance-type programs: general / special management-stabilisation funds, temporary-distress and disaster support, restart funds, restructuring funds, business-conversion funds. Example domestic-market programs: direct-production confirmation, public-procurement support, market development.

CORRELATION READING

"Support lowers growth"

Beneficiary statistics, raw difference. Selection and effect are indistinguishable.

CAUSAL READING

"Selection dominates"

Balanced pre-trends, persistent post-gap, low matching success, 89% crisis-response share.

DECISION

Use, qualify, extend

Rely on low-distortion groups (e.g. smart manufacturing & AI); collect rejected-applicant data for the finance group.

Figure 14 – Same data, three conclusions. Only the causal reading turns an apparently paradoxical result into a defensible policy statement.

FROM ESTIMATE TO DECISION: SIX MENUS

Causal estimates create value only when they reach the people who decide. In the PoC, all users are public officials – firms do not log in themselves (confirmed with KOSI on 9 July 2026). The challenge required four scenarios; we organised the service around job roles and delivered six menus (Figure 15), all sharing one five-step engine: condition input → search for similar targets → causal inference → quantification → report.

The interface uses business language instead of statistical jargon (“expected effect for this firm”), shows the evidence next to each number (“23 similar firms grew sales by 15%”) and labels every value with an A / B / C confidence grade – weak evidence is displayed, not hidden. Screen and PDF report render from a single data source, so the number an official sees is the number that is filed.

Figure 16 summarises the measured performance. Menu 5 adds a budget view: the Herfindahl–Hirschman index (HHI) and Gini coefficient measure how concentrated support is, and incremental sales and employment per unit of budget measure efficiency.

MENU 1 · BUSINESS SUPPORT
Firm-specific program finder
“Which programs fit this firm?”
3+ recommended programs with expected effect · CATE / GATE

MENU 2 · FIELD DISCOVERY
Unsupported-firm discovery
“What if this firm participated?”
Real-time CATE, auto-generated proposal · GATE

MENU 3 · ALL OFFICIALS
Integrated document search
“Find the related notices.”
RAG over notices and source documents with citations

MENU 4 · POLICY EVALUATION
Ex-post effect analysis
“Did this program work?”
Beneficiary vs. matched non-beneficiary, Y0 / Y1 · ATT

MENU 5 · DESIGN & BUDGET
Distribution & efficiency
“Is the budget spread well?”
HHI, Gini, effect per unit of budget

MENU 6 · DESIGN & BUDGET
Ex-ante prediction
“What will a new program achieve?”
Counterfactual simulation from target group, size, duration · GATE

Figure 15 – Six role-based menus covering the four required functions: policy matching, effect analysis, effect prediction and automatic report generation.

Indicator	Challenge criterion	EVA target	Measured in PoC	Status
CATE error rate	none (self-defined)	≤ 15%	2.6–11.9%, 8/8 combinations	**PASS**
DoWhy refutation	none (self-defined)	≥ 3 / 4	4 / 4, both cohorts	**PASS**
Recommendation fit	≥ 80%	≥ 85%	100% (expert review, knowledge-graph data)	**PASS**
Report satisfaction	≥ 80 points	≥ 85 points	87.5 points (team-defined criterion)	**PASS**
Golden-set quality	70%	90%+	80–100% (team-defined criterion)	**PASS***
System response time	≤ 10 s	≤ 10 s	Menus 1, 2, 4, 6 met; menus 3, 5 exceed	**Partial**

Figure 16 – Performance indicators, targets and measured results (GPU server, live measurement). Menus 3 and 5 accumulate retrieval, LLM summarisation and multiple subgroup estimates within one query; their processing scope is stated on screen. *Golden-set quality exceeds the 70% challenge criterion; EVA’s own 90% target is reached in the upper part of the measured range.

WHY FUTURE POLICY AI MUST BE CAUSAL

Large language models have made it easy to produce fluent policy reports. They have not made it easier to know whether a program works. A model that summarises beneficiary statistics in perfect prose still reports a correlation – only more convincingly. The more persuasive AI-generated analysis becomes, the more important it is that the numbers underneath are causal, tested and traceable.

The correlation between participation and outcome can be computed from two columns. As every analyst knows, however, correlation does not equal causation. In a sense, causation is the opposite of correlation: correlation needs only the two fields in question, while causation needs *everything else* about the firm – its industry, size, region, history, finances and trend – because each of these is a candidate common cause. Only when no confounder can explain an observed association may we suspect a direct effect.

This is why we believe four properties will define the next generation of policy AI:

1. **Interventional questions.** The system answers `P(Y | do(T))`, not `P(Y | T)`, and makes its identifying assumptions explicit in a DAG that can be reviewed.
2. **Heterogeneous answers.** Policy is targeted; a single average hides the firms for which a program works and those for which it does not. CATE turns "does it work?" into "for whom does it work, and by how much?"
3. **Self-falsification.** Models are attacked by refutation tests before release, and headline estimates are replicated on an independent cohort.
4. **Computed, not generated, numbers.** Language models explain; rule-based causal engines decide the figures. Self-hosted models keep firm data away from external AI providers.

WHAT THE POC CANNOT YET IDENTIFY – AND WHAT COMES NEXT

The case study shows the boundary of observational data: confounders W1–W9 do not fully remove selection into crisis-response support. Three data extensions address this directly; the data enter the same pipeline, and the screening-score case adds a regression-discontinuity estimator:

- **Rejected applicants as controls.** Firms that applied but were not selected share the decision to apply; the remaining difference narrows to the screening criteria.
- **Longer pre-treatment trends.** Three or more years of history allow a direct test of the parallel-trend assumption that underpins quasi-experimental designs.
- **Screening scores and cut-offs.** Where a score threshold exists, a regression discontinuity design compares firms just above and below it – close to a randomised experiment.

The structure "participation (T) → outcome (Y), given characteristics (W)" is the same in employment, welfare and industrial policy. The same pipeline can therefore be transferred to other policy domains and institutions by adding data, not by redevelopment.

Understanding potential cause-and-effect relationships requires a holistic, causal view of the program in focus. The analysis of real-world policy and future AI decision systems will need exactly that. Causal Policy Analytics constitutes a tremendous opportunity to make public spending accountable – not by showing more numbers, but by showing the right ones.

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Data note. All PoC figures in this paper are taken from references 8 and 9 and refer to the data KOSI provided for the challenge: firm basic and financial records (7,921,066 rows), support history (4.42 million records), program information (210 programs), business notices (833) and the 2024 and 2025 support-policy satisfaction surveys. Firm identifiers were de-identified by the data provider at source; all derived outputs remain de-identified. Effects are reported in percentage points (%p) as in the source analysis.

Technology note. Document parsing: pdfplumber, PyMuPDF, olefile + lxml (HWP/HWPX), Gemma 4 multimodal OCR for image-based notices. Retrieval: ChromaDB with granite-embedding (278M). Knowledge graph: Neo4j. Causal core: DoWhy, EconML (CPU). Language model: Gemma 4 31B via Ollama, self-hosted on a dedicated GPU server (NHN Cloud H100); no external LLM API in normal operation. Interfaces: FastAPI REST and Streamlit, served over HTTPS through an outbound-only Cloudflare Tunnel.

EVA's approach is the subject of Korean patent applications No. 2026-0094355 (filed 26 May 2026; automatic AST structuring and canonical JSON storage of multi-format documents using function-separated small language models) and No. 2026-0060871 (filed 3 April 2026; general-purpose AI decision-support system integrating ontology-based causal graph inference, GraphRAG-LLM question answering and feedback learning).



Decision intelligence, powered by causality.

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